



JABchem



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Past Papers Nat 5 Chemistry

2019 Marking Scheme

Grade Awarded	Mark Required		% candidates achieving grade
	(/125)	%	
A	89+	71.2%	34.6%
B	74+	59.2%	21.6%
C	59+	47.2%	20.7%
D	44+	35.2%	15.9%
No award	<44	<35.2%	7.2%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	17.2 /25	41.5 /75	18.2 /25

2019 National 5 Chemistry Marking Scheme

MC Qu	Answer	% Pupils Correct	Reasoning												
1	C	87	$\text{Rate} = \frac{\Delta \text{Quantity}}{\Delta \text{Time}} = \frac{5 \text{ cm}^3}{2 \text{ min}} = 2.5 \text{ cm}^3 \text{ min}^{-1}$												
2	B	64													
3	C	90	☒ A All elements in group 1 have 1 electron in outer shell ☒ B All elements in group 2 have 2 electrons in outer shell ☒ C All elements in group 7 have 7 electrons in outer shell ☒ D All elements in group 8 (apart from Helium) have 8 electrons in outer shell												
4	D	72	☒ A This ion has a negative charge as it has more electrons than protons ☒ B This atom has no charge as it has equal numbers of protons and electrons ☒ C This atom has no charge as it has equal numbers of protons and electrons ☒ D This ion has a 2+ positive charge as it has 2 less electrons than protons												
5	B	76	☒ A 40°C is above the critical temperature to CO ₂ is a gas at 40°C ☒ B NH ₃ and CO ₂ are compounds and have the highest critical temperatures ☒ C Ammonia NH ₃ has the highest critical temperature but is medium in mass ☒ D Hydrogen H ₂ is has a lower critical temperature than the noble gas Helium He												
6	D	80	<table border="1"> <thead> <tr> <th>HF</th><th>F₂O</th><th>CF₄</th><th>PF₃</th></tr> </thead> <tbody> <tr> <td></td><td></td><td></td><td></td></tr> <tr> <td>Linear</td><td>Angular</td><td>Tetrahedral</td><td>Trigonal Pyramidal</td></tr> </tbody> </table>	HF	F ₂ O	CF ₄	PF ₃					Linear	Angular	Tetrahedral	Trigonal Pyramidal
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Linear	Angular	Tetrahedral	Trigonal Pyramidal												
7	A	45	☒ A Na ⁺ electron arrangement is 2,8 and O ²⁻ electron arrangement is 2,8 ☒ B Li ⁺ electron arrangement is 2 and F ⁻ electron arrangement is 2,8 ☒ C K ⁺ electron arrangement is 2,8,8 and Br ⁻ electron arrangement is 2,8,18,8 ☒ D Mg ²⁺ electron arrangement is 2,8 and Cl ⁻ electron arrangement is 2,8,8												
8	B	66	☒ A ionic copper sulphate in the solid state is a non-conductor ☒ B ionic compounds will conduct is both the molten/liquid state and in solution ☒ C ionic potassium nitrate in the solid state is a non-conductor ☒ D hexane C ₆ H ₁₄ is a hydrocarbon and is covalent molecular and is a non-conductor												
9	A	58	☒ A Calcium hydroxide is the solute as it is the solid which is dissolved ☒ B Water is the solvent as it is the liquid doing the dissolving ☒ C Calcium hydroxide solution is the solution with the solute dissolved in solvent ☒ D Calcium hydroxide must be soluble if it dissolves												
10	B	73	$\text{gfm NH}_4\text{NO}_3 = (1 \times 14) + (4 \times 1) + (1 \times 14) + (3 \times 16) = 14 + 4 + 14 + 48 = 80$ $\% \text{N} = \frac{\text{mass of nitrogen}}{\text{gfm}} \times 100 = \frac{28}{80} \times 100 = 35\%$												
11	D	78	☒ A The pH of an alkaline solution will decrease to pH=7 on dilution ☒ B The pH of an alkaline solution will decrease to pH=7 on dilution ☒ C The concentration of OH ⁻ ions decreases on dilution ☒ D The concentration of OH ⁻ ions decreases on dilution												

12	A	57	☒ A sodium oxide is a metal oxide (a type of base) and neutralises an acid to form water ☒ B calcium chloride does not react with acids and is not a base ☒ C potassium nitrate does not react with acids and is not a base ☒ D ammonium sulphate does not react with acids and is not a base																																																														
13	D	65	☒ A Cyclopropane C_3H_6 has an isomer called propene C_3H_6 ☒ B But-1-ene C_4H_8 has isomers including but-2-ene C_4H_8 ☒ C Pentane C_5H_{12} has isomers including 2-methylbutane C_5H_{12} . ☒ D Ethene C_2H_4 has no isomers																																																														
14	C	63	☒ A Formula is C_6H_{12} so does not fit the general formula of alkanes C_nH_{2n+2} ☒ B Formula is C_6H_{12} so does not fit the general formula of alkanes C_nH_{2n+2} ☒ C C=C double bond between C_2 & C_3 (numbered from right) and methyl group on C_3 ☒ D C=C takes the lower number system so Pent-3-ene should be pent-2-ene																																																														
15	A	39	☒ A Oct-2-ene produces two products on hydration (octan-2-ol and octan-3-ol) ☒ B Hex-3-ene produces one product on hydration (hexan-3-ol) ☒ C But-2-ene produces one product on hydration (butan-2-ol) ☒ D Ethene produces one product on hydration (ethanol)																																																														
16	B	88	Reaction removes the C=O group and replaces it with a CH_2 group. Alkane produced has 6 carbons in main chain and a methyl group on C_2 $\therefore$ 2-methylhexane produced																																																														
17	B	65	<table><tr><td>Carboxylic acid</td><td>Methanoic Acid</td><td>Ethanoic Acid</td><td>Propanoic Acid</td><td>Butanoic Acid</td><td>Pentanoic Acid</td><td>Hexanoic Acid</td><td>Heptanoic Acid</td><td>Octanoic Acid</td></tr><tr><td>Formula</td><td>HCOOH</td><td>CH₃COOH</td><td>C₂H₅COOH</td><td>C₃H₇COOH</td><td>C₄H₉COOH</td><td>C₅H₁₁COOH</td><td>C₆H₁₃COOH</td><td>C₇H₁₅COOH</td></tr><tr><td>Mass</td><td>low</td><td></td><td></td><td></td><td></td><td></td><td></td><td>high</td></tr><tr><td>Melting Point</td><td>low</td><td></td><td></td><td></td><td></td><td></td><td></td><td>high</td></tr><tr><td>Boiling Point</td><td>low</td><td></td><td></td><td></td><td></td><td></td><td></td><td>high</td></tr><tr><td>Solubility</td><td>high</td><td></td><td></td><td></td><td></td><td></td><td></td><td>low</td></tr></table>	Carboxylic acid	Methanoic Acid	Ethanoic Acid	Propanoic Acid	Butanoic Acid	Pentanoic Acid	Hexanoic Acid	Heptanoic Acid	Octanoic Acid	Formula	HCOOH	CH ₃ COOH	C ₂ H ₅ COOH	C ₃ H ₇ COOH	C ₄ H ₉ COOH	C ₅ H ₁₁ COOH	C ₆ H ₁₃ COOH	C ₇ H ₁₅ COOH	Mass	low							high	Melting Point	low							high	Boiling Point	low							high	Solubility	high							low								
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18	D	82	☒ A Largest voltage = largest separation on electrochemical series (magnesium-silver) ☒ B 2 nd Largest voltage = 2 nd largest separation on electrochemical series (zinc-silver) ☒ C 3 rd Largest voltage = 3 rd largest separation on electrochemical series (iron-silver) ☒ D Smallest voltage = smallest separation on electrochemical series (copper-silver)																																																														
19	C	66	☒ A Metal Z is more reactive than metal X as metals Z reacts with water ☒ B Metal Y is least reactive metal as it is only one which had no reaction with acid ☒ C Metal Y is least reactive metal and Metal Z is most reactive metal ☒ D Metal Y is least reactive metal as it is only one which had no reaction with acid																																																														
20	C	47	<table><tr><td>$\begin{array}{c} CH_3H \\ \quad \\ C=C \\ \quad \\ H \quad CH_3 \end{array}$</td><td>$\begin{array}{c} CH_3H \\ \quad \\ C=C \\ \quad \\ H \quad H \end{array}$</td><td colspan="7">$\begin{array}{ccccccc} CH_3H & CH_3H & CH_3H & CH_3H & CH_3H & CH_3H & CH_3H \\ & & & & & & \\ -C- & C- & C- & C- & C- & C- & C- \\ & & & & & & \\ H & CH_3 & H & H & H & H & CH_3 \end{array}$</td></tr><tr><td colspan="2"></td><td></td><td>Monomer 1</td><td>Monomer 2</td><td colspan="4">Co-polymer</td><td colspan="2"></td></tr><tr><td>21</td><td>C</td><td>29</td><td colspan="9">☒ A Nitrogen dioxide is soluble and would dissolve in water rather than collect. ☒ B Nitrogen dioxide is soluble and would dissolve in water rather than collect. ☒ C Nitrogen dioxide is more dense than air and would collect in bottom of cylinder ☒ D Nitrogen dioxide is more dense than air and would escape out of the cylinder</td></tr><tr><td>22</td><td>A</td><td>95</td><td colspan="9">☒ A Ba^{2+} ions give a green colour in a flame test ☒ B Ca^{2+} ions give a orange-red colour in a flame test ☒ C K^+ ions give a lilac colour in a flame test ☒ D Na^+ ions give a yellow colour in a flame test</td></tr><tr><td>23</td><td>D</td><td>84</td><td colspan="9">☒ A oxygen gas relights a glowing splint ☒ B nitrogen does not turn lime water milky/burn with a pop/relight a glowing splint ☒ C hydrogen gas burns with a pop ☒ D carbon dioxide turns limewater milky</td></tr></table>	$\begin{array}{c} CH_3H \\ \quad \\ C=C \\ \quad \\ H \quad CH_3 \end{array}$	$\begin{array}{c} CH_3H \\ \quad \\ C=C \\ \quad \\ H \quad H \end{array}$	$\begin{array}{ccccccc} CH_3H & CH_3H & CH_3H & CH_3H & CH_3H & CH_3H & CH_3H \\ & & & & & & \\ -C- & C- & C- & C- & C- & C- & C- \\ & & & & & & \\ H & CH_3 & H & H & H & H & CH_3 \end{array}$										Monomer 1	Monomer 2	Co-polymer						21	C	29	☒ A Nitrogen dioxide is soluble and would dissolve in water rather than collect. ☒ B Nitrogen dioxide is soluble and would dissolve in water rather than collect. ☒ C Nitrogen dioxide is more dense than air and would collect in bottom of cylinder ☒ D Nitrogen dioxide is more dense than air and would escape out of the cylinder									22	A	95	☒ A Ba^{2+} ions give a green colour in a flame test ☒ B Ca^{2+} ions give a orange-red colour in a flame test ☒ C K^+ ions give a lilac colour in a flame test ☒ D Na^+ ions give a yellow colour in a flame test									23	D	84	☒ A oxygen gas relights a glowing splint ☒ B nitrogen does not turn lime water milky/burn with a pop/relight a glowing splint ☒ C hydrogen gas burns with a pop ☒ D carbon dioxide turns limewater milky														
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24	C	66	☒ A Addition reactions happen when a molecule adds across as C=C double bond ☒ B Combustion reactions happen when chemicals split up and react with oxygen ☒ C Precipitation reactions form an insoluble solid when solutions are added together ☒ D Neutralisation reactions happen when an acid reacts with a base to form water
25	B	77	$\text{AgNO}_3(\text{aq}) + \text{NaBr}(\text{aq}) \rightarrow \text{NaNO}_3(\text{aq}) + \text{AgBr}(\text{s})$ Split solutions into ions $\text{Ag}(\text{aq}) + \text{NO}_3^-(\text{aq}) + \text{Na}^+(\text{aq}) + \text{Br}^-(\text{aq}) \rightarrow \text{Na}^+(\text{aq}) + \text{NO}_3^-(\text{aq}) + \text{AgBr}(\text{s})$ Identify Spectator Ions and cancel out Spectator Ions $\text{Ag}(\text{aq}) + \cancel{\text{NO}_3^-(\text{aq})} + \cancel{\text{Na}^+(\text{aq})} + \text{Br}^-(\text{aq}) \rightarrow \cancel{\text{Na}^+(\text{aq})} + \cancel{\text{NO}_3^-(\text{aq})} + \text{AgBr}(\text{s})$ Re-write equation without spectator ions $\text{Ag}(\text{aq}) + \text{Br}^-(\text{aq}) \rightarrow \text{AgBr}(\text{s})$

2019 National 5 Chemistry Marking Scheme

Long Qu	Answer	Reasoning								
1a	network	SiO ₂ is a covalent compound as both elements in the compound are non-metals. If the covalent compound has a high melting point it is a covalent network compounds. If it has a low melting and boiling point then it is covalent molecular.								
1b(i)	Isotope	Same Atomic number Number of protons Different Mass number Number of neutrons								
1b(ii)	Different number of neutrons	Both isotopes of boron have 5 protons as they have an atomic number of 5. The ¹⁰ B isotope has 5 neutrons and the ¹¹ B isotope has 6 neutrons, calculated by number of neutrons = mass number - atomic number								
1b(iii)	More ¹¹ B isotope	The RAM (Relative Atomic Mass) is the average mass of the different isotopes. As the average is much closer to 11, there must be more ¹¹ B isotope in the sample.								
1c	120	24% of 500g = $\frac{24}{100} \times 500 = 120\text{g}$								
2a	Hydroxyl	<table><tr><td>—O—H</td><td>$\begin{array}{c} \text{O} \\ \\ \text{—C—OH} \end{array}$</td></tr><tr><td>hydroxyl group</td><td>carboxyl group</td></tr></table>	—O—H	$\begin{array}{c} \text{O} \\ \\ \text{—C—OH} \end{array}$	hydroxyl group	carboxyl group				
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hydroxyl group	carboxyl group									
2b	enzyme	Problem Solving: Retrieving information from a passage								
2c	calcium oxalate	Acids react with bases to form salts. Oxalic acid reacts with calcium ions to form the salt calcium oxalate								
2d	90	Lactic acid is the harmless product mentioned in the text. Formula of lactic acid = C ₃ H ₆ O ₃ 1 mol = (3x12) + (6x1) + (3x16) = 36 + 6 + 48 = 90								
3a	One diagram from:									
3b	Blue or purple	Ammonia NH ₃ gas dissolves in water to form alkaline ammonium hydroxide solution. Alkali on the damp/moist pH paper will turn blue/purple. $\text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l}) \longrightarrow \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$ ammoniawaterammonium ionhydroxide ion								
3c(i)	Increase in temperature decrease in percentage	<table><tr><td>One From:</td><td>As the temperature increases yield/% decreases.</td><td>As the temperature decreases the yield/% increases.</td><td>The yield/% increases as the temperature decreases.</td><td>The yield/% decreases as the temperature increases.</td></tr></table> Cause and Effect: It must be a change of temperature and its effect on yield so "as yield of ammonia decreases, temperature gets higher" is incorrect	One From:	As the temperature increases yield/% decreases.	As the temperature decreases the yield/% increases.	The yield/% increases as the temperature decreases.	The yield/% decreases as the temperature increases.			
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3c(ii)	Graph showing:	<table><tr><td>1 mark</td><td>1 mark</td><td>1 mark</td><td>1 mark</td></tr><tr><td>One mark is awarded for a graph which shows points plotted rather than bars.</td><td>The axis/axes of the graph has/have suitable scale(s). - plotted points occupies at least half of the width and half of the height of the graph paper - The axes have suitable scales </td><td>The axes of the graph have suitable labels and units.</td><td>All data points plotted accurately (within a half box tolerance) with either a line of best fit drawn or plots joined by curve. This mark can only be accessed if linear scales for both axes have been provided.</td></tr></table>	1 mark	1 mark	1 mark	1 mark	One mark is awarded for a graph which shows points plotted rather than bars.	The axis/axes of the graph has/have suitable scale(s). - plotted points occupies at least half of the width and half of the height of the graph paper - The axes have suitable scales	The axes of the graph have suitable labels and units.	All data points plotted accurately (within a half box tolerance) with either a line of best fit drawn or plots joined by curve. This mark can only be accessed if linear scales for both axes have been provided.
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3d(i)	Electrolysis and lithium nitride	Processes are in the circular boxes. Electrolysis is the circular box answer Chemicals are in the square boxes. Lithium nitride is formed from lithium and nitrogen.								

3d(ii)	One arrow drawn from:	water water or lithium hydroxide lithium hydroxide																												
4a	nucleus	All nuclear reactions take place in the nucleus.																												
4b(i)	One answer from:	The time for half of the mass to decay The time for half of the (radio)activity to decay The time for half of the nuclei to decay																												
4b(ii)	87.5%	<table><tr><th>Time (days)</th><th>Percentage Remaining</th></tr><tr><td>0</td><td>100%</td></tr><tr><td>8</td><td>50%</td></tr><tr><td>16</td><td>25%</td></tr><tr><td>24</td><td>12.5%</td></tr></table> 12.5% remaining after 24 days 87.5% must have decayed by 24 days					Time (days)	Percentage Remaining	0	100%	8	50%	16	25%	24	12.5%														
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4b(iii)	Stays the same	Half-life is independent of concentration, temperature and state of matter (solid, liquid, gas or solution)																												
5a	Bromine decolourises	Bromine adds across C=C double bonds and bromine is decolourised in the process.																												
5b(i)	Hydrogenation	The addition of hydrogen across C=C double bond is known as hydrogenation. The addition of water across C=C double bond is known as hydration.																												
5b(ii)	Chlorine	Two chlorine atoms have joined across the location of where the C=C double bond used to be. Cl ₂ is the reactant in formation of compound Y (1,2-dichloropropane)																												
5b(iii)	Poly(propene)	If propene is the monomer then the polymer is poly(propene)																												
5c(i)	140	1mol C₆H₁₀ = (6x12) + (10x1) = 72 + 10 = 82g no. of mol = $\frac{\text{mass}}{\text{gfm}}$ = $\frac{410\text{g}}{82\text{g mol}^{-1}}$ = 5mol C₂H₄ + C₄H₆ —————> C₆H₁₀ 1mol 5mol 1mol 5mol 1mol C₂H₄ = (2x12) + (4x1) = 24 + 4 = 28g mass = no. of mol x gfm = 5 mol x 28g mol⁻¹ = 140g																												
5c(ii)	Answer to include:	1 mark: Cyclopentene has weaker (forces of attraction) 1 mark: Forces between molecules/intermolecular attractions mentioned																												
6	Open Question:	<table><tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr><tr><td>Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>					3 mark answer	2 mark answer	1 mark answer	Demonstrates a good understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a reasonable understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a limited understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.																		
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7a	Same/similar chemical properties <u>and</u> same general formula.	Members of the same homologous series have the same/similar properties e.g. all alkenes will decolourise bromine solution quickly. The general formula must fit all members of the homologous series e.g. all alkanes fit the general formula C _n H _{2n+2} e.g. methane CH ₄ , ethane C ₂ H ₆ , propane C ₃ H ₈ .																												
7b(i)	Alkane	The hydrocarbon C ₂₅ H ₅₂ must belong to the alkane family as it fits the general formula of alkanes of C _n H _{2n+2} . In n=35 then 2n+2 = (2x25) +2 = 50+2 = 52 ∴ C ₂₅ H ₅₂																												
7b(ii)	C ₃₅ H ₇₂	General Formula C _n H _{2n+2} where number of H atoms = 72. Hydrogen number 2n+2 = 72 ∴ 2n = 70 ∴ n=35 ∴ formula = C ₃₅ H ₇₂																												
7c	391-394 (inclusive)	<table><tr><td>No. of Carbons</td><td>20</td><td>21</td><td>22</td><td>23</td><td>24</td></tr><tr><td>Boiling Point (°C)</td><td>343</td><td>356</td><td>369</td><td>381</td><td>-</td></tr><tr><td>Difference:</td><td></td><td>12</td><td>13</td><td>13</td><td>Predicted 13</td></tr><tr><td>Prediction:</td><td>-</td><td>-</td><td>-</td><td>-</td><td>394</td></tr></table>					No. of Carbons	20	21	22	23	24	Boiling Point (°C)	343	356	369	381	-	Difference:		12	13	13	Predicted 13	Prediction:	-	-	-	-	394
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Prediction:	-	-	-	-	394																									

8a	Supernova	Problem Solving: Gathering information from a passage					
8b	beryllium aluminium silicon oxygen	Beryl is beryllium aluminium silicate. Silicate means both silicon and oxygen are also found in the compound.					
8c	$\text{BeCl}_2 + \text{K} \rightarrow \text{Be} + \text{KCl}$	Beryllium chloride + Potassium $\rightarrow$ Beryllium + Potassium chloride BeCl_2 + K $\rightarrow$ Be + KCl BeCl_2 + 2K $\rightarrow$ Be + 2KCl					
8d	Reduction	Beryllium ions					

10b(iii)	$2\text{Fe}^{3+} + 2\text{I}^{-}$ $\downarrow$ $2\text{Fe}^{2+} + \text{I}_2$	$\begin{array}{rcll} 2\text{I}^{-} & \rightarrow & \text{I}_2 & + 2\text{e}^{-} \\ \text{Fe}^{3+} + \text{e}^{-} & \rightarrow & \text{Fe}^{2+} & \end{array}$ Multiple equations to get same number of electrons $\begin{array}{rcll} 2\text{I}^{-} & \rightarrow & \text{I}_2 & + 2\text{e}^{-} \\ 2\text{Fe}^{3+} + 2\text{e}^{-} & \rightarrow & 2\text{Fe}^{2+} & \end{array}$ Cancel out electrons add equations together $2\text{Fe}^{3+} + 2\text{I}^{-} \rightarrow 2\text{Fe}^{2+} + \text{I}_2$			
10c	Conducts electricity	Graphite has delocalised electrons that do not take part in the covalent bonding and are free to jump from atom to atom to allow electrical conduction.			
11a	$\text{Ca}^{2+}\text{CO}_3^{2-}$	Write down Valency below each element's symbol $\begin{array}{cc} \text{Ca} & \text{CO}_3 \\ 2 & 2 \end{array}$	Put in Cross-over Arrows $\begin{array}{cc} \text{Ca} & \text{CO}_3 \\ 2 & 2 \end{array}$	Follow arrows to get formula $\text{Ca}_2(\text{CO}_3)_2$ Cancel Down CaCO_3	
11b	propanoic acid	$\begin{array}{rcll} \text{acid} & + & \text{metal carbonate} & \rightarrow \text{salt} + \text{water} + \text{carbon dioxide} \\ \text{propanoic acid} & + & \text{calcium carbonate} & \rightarrow \text{calcium propanoate} + \text{water} + \text{carbon dioxide} \end{array}$			
11c	One answer from:	no more solid reacts/ until it no longer reacts calcium carbonate left (at the bottom)	solid remains/is left (at bottom of the beaker) no more calcium carbonate reacts	a gas is no longer produced no more fizzing/bubbling neutral/neutralised with a description of testing pH.	
11d	Filtration Evaporation	Filtration is needed to separate the unreacted calcium carbonate from the products. The residue of filtration is calcium propanoate solution. The solid propanoate is recovered by evaporating the water in the solution using an evaporating basin and a Bunsen burner.			
12a(i)	standard	A standard solution is a solution whose concentration is accurately know. It can then be used to work out the concentration of another solution with which it reacts with.			
12a(ii)	21.2	$\text{no. of mol} = \text{volume} \times \text{concentration} = 0.2\text{litres} \times 1\text{mol l}^{-1} = 0.2\text{mol}$ $\text{gfm Na}_2\text{CO}_3 = (2 \times 23) + (1 \times 12) + (3 \times 16) = 46 + 12 + 48 = 106\text{g mol}^{-1}$ $\text{mass} = \text{no of mol} \times \text{gfm} = 0.2\text{mol} \times 106\text{g mol}^{-1} = 21.2\text{g}$			
12b(i)	One answer from:	Move burette/ sodium carbonate (reading) to eye level	Reduce distance between flask and burette	Use white tile/paper	Clamp the burette towards the centre
12b(ii)	One answer from:	To show the end-point	To show point of neutralisation	To show completion of the reaction	
12b(iii)	concordant	Titrations using a burette allow exact volumes of solutions to be worked out. After the initial titration to allow a rough volume for the reaction to be found, the experiment is repeated exactly in a more accurate way to work out the exact volume of solution needed to completely react. This is repeated until two volumes within $\pm 0.2\text{cm}^3$ of each other are achieved. These are described as concordant. The concordant volumes are averaged and the rough titre is ignored.			
12b(iv)	1.5	$\text{no. of mol Na}_2\text{CO}_3 = \text{volume} \times \text{concentration} = 0.015\text{litres} \times 1\text{mol l}^{-1} = 0.015\text{mol}$ $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$ $\begin{array}{ccc} 1\text{mol} & 2\text{mol} & \\ 0.015\text{mol} & 0.030\text{mol} & \end{array}$ $\text{concentration} = \frac{\text{no of moles}}{\text{volume}} = \frac{0.030\text{mol}}{0.02\text{litres}} = 1.5\text{ mol l}^{-1}$			
13	Open Question:	3 mark answer Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	2 mark answer Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	1 mark answer Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.	

2019 National 5 Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	C	1	10	87	1	0	0
2	B	31	64	3	1	0	0
3	C	2	4	90	3	0	0
4	D	9	12	7	72	0	0
5	B	8	76	6	10	0	0
6	D	1	1	18	80	0	0
7	A	45	23	12	18	1	0
8	B	10	66	12	12	0	0
9	A	58	28	12	2	0	0
10	B	20	73	3	3	0	0
11	D	5	3	13	78	0	0
12	A	57	17	14	12	0	0
13	D	15	11	9	65	0	0
14	C	21	5	63	11	0	0
15	A	39	17	23	20	1	0
16	B	4	88	5	3	0	0
17	B	15	65	16	4	0	0
18	D	6	3	8	82	0	0
19	C	2	31	66	1	0	0
20	C	17	11	47	25	1	0
21	C	12	49	29	9	0	0
22	A	95	1	2	2	0	0
23	D	2	7	7	84	0	0
24	C	16	6	66	12	0	0
25	B	8	77	11	4	0	0